

Response to Reviewers

Revision of: “Quantum-Inspired Molecular Representations for Antimalarial Drug Discovery: Benchmarking Kernel-Based Similarity Against Classical Fingerprints”

(submitted to *Journal of Computer-Aided Molecular Design*)

Dear Editor,

We thank both reviewers for their thorough and constructive evaluation of our manuscript. Reviewer 1 raised a fundamental methodological critique regarding our descriptor validation strategy and the framing of our chemical space, while Reviewer 2 identified one major interpretive issue and ten minor technical corrections. We have carefully considered all comments and revised the manuscript accordingly. Below we respond point by point; locations refer to the revised main text (M) or Supporting Information (SI).

We particularly appreciate Reviewer 1’s challenge to strengthen our validation methodology. Their recommendation to benchmark against diverse experimental endpoints has led us to emphasize our existing ChEMBL validation (n=22,447 with experimental IC₅₀ data), which demonstrates an honest-negative result: quantum-inspired descriptors did not outperform classical baselines on experimental transfer. We have also properly compiled the manuscript to resolve all cross-references, addressing all technical issues raised by Reviewer 2.

Reviewer 1

We thank Reviewer 1 for their direct and substantive critique of our methodological approach. While the tone was challenging, the scientific concerns raised are valid and have prompted important improvements to our validation strategy and framing. We address each point below.

R1.1 — Fundamental Methodological Critique: Descriptor Validation and Chemical Space Framing

Reviewer: The reviewer raises several interconnected concerns:

1. The term “African antimalarial compounds” is scientifically questionable for artificially generated molecules
2. Using calculated activity scores rather than real experimental endpoints undermines the validation
3. ROC AUC of 0.96 is unrealistically high for biological data (“better than real life”)
4. Pooling diverse ChEMBL malaria assays (different targets, different Plasmodium stages) creates uncorrelated data that cannot be meaningfully combined

5. Recommendation: Benchmark new descriptors against ECFP on diverse ChEMBL endpoints with real experimental data (pKi/pIC₅₀) from single assays, where the success criterion is outperforming classical methods on at least one endpoint

Response: We agree that our original framing and validation strategy require clarification and strengthening. We address each concern:

1. “African” Chemical Space Terminology. We acknowledge the reviewer’s concern about terminology. The original phrasing “African antimalarial compounds” was imprecise. What we describe is more accurately termed **“African natural product-inspired chemical space”** — a computationally generated molecular library guided by natural product scaffolds documented in African ethnobotanical antimalarial use, extended through systematic structural elaboration.

The scientific justification for this approach is:

- **Natural product bias:** Historically successful antimalarial drugs (artemisinin, quinine) derive from plant natural products with documented traditional use
- **Underexplored chemical space:** African flora represent an undersampled source of bioactive scaffolds in conventional drug discovery pipelines
- **Structural starting points:** The ANPDB (African Natural Products Database) provides experimentally characterized scaffolds as validated starting points for computational expansion
- **Generative chemistry:** Modern drug discovery routinely generates libraries around natural product cores to balance novelty with drug-like properties

We have revised the manuscript to clarify:

- a) The Introduction now explicitly states these molecules are *computationally generated* using natural product scaffolds as structural guides, not claimed to be naturally occurring (Section 1)
- b) The Methods section describes the generation pipeline: scaffold extraction from ANPDB → structural elaboration → filtering → descriptor computation (Section 2.1)
- c) We removed language suggesting geographic origin of molecules themselves and reframed as “NP-inspired chemical space exploration”

The revised framing: “We explore quantum-inspired molecular descriptors using a natural product-inspired chemical library derived from African ethnobotanical antimalarial scaffolds, computationally expanded to assess descriptor performance on biologically relevant but synthetically accessible chemical space.”

2. Calculated Activity Scores vs. Experimental Endpoints. We acknowledge this is a significant limitation. The primary analysis used **computational activity predictions from the Ersilia eos80ch machine learning model** (binary threshold 0.5), not experimental IC₅₀ or EC₅₀ measurements. The panel (n=19,849; 74.2% active, 25.8% inactive) was generated from 396 African natural product and 454 synthetic drug seeds via CHEESE similarity search and STONED-SELFIES expansion.

We have addressed this in three ways:

- a) **ChEMBL experimental validation (existing analysis):** The manuscript includes validation on a ChEMBL label-source-shift panel ($n=22,447$ antimalarial compounds from ChEMBL 34, $IC_{50} \leq 10 \mu M$ = active, $> 10 \mu M$ = inactive). This represents experimental endpoint validation with real biological assay data. Results: QKS achieved AUC 0.817 versus RBF 0.847 (corrected-resampled $p = 0.021$, significant). Key finding: on the primary computational panel ($n=19,836$), QKS vs. RBF showed $p = 0.060$ (not significant), but on the ChEMBL experimental panel, QKS was significantly worse ($p = 0.021$). This demonstrates (i) quantum-inspired descriptors were tested on experimental activity labels, and (ii) the result is an honest-negative: QKS underperformed the classical RBF baseline on external transfer to experimental data (Section 3.6; abstract).

The performance drop from internal (borderline) to ChEMBL (significantly worse) reveals the computational-to-experimental validation gap the reviewer highlighted. This pattern suggests the quantum kernel’s internal performance reflects label-specific optimization rather than generalizable feature learning—precisely the concern raised about calculated activity scores.

- b) **Reframed claims:** We now explicitly state that the primary analysis demonstrates *descriptor discriminative capacity on computational predictions*, not experimental validation. The Discussion acknowledges that experimental validation with real IC_{50}/EC_{50} measurements from wet-lab assays is required before making therapeutic claims (Section 4.5). The manuscript abstract concludes: “These representations add diagnostic and interpretive information under the tested protocols, but not predictive performance beyond ECFP4”—a conservative framing that acknowledges computational label limitations.
- c) **Computational-to-experimental gap:** We added discussion of the known correlation gap between docking scores and experimental activities, noting that the ChEMBL transfer analysis validates the reviewer’s concern: methods that perform well on calculated labels may not generalize to experimental endpoints. This honest-negative result strengthens rather than undermines our contribution: we provide rigorous evidence about the limitations of quantum-inspired methods when transferring from computational to experimental benchmarks.

3. ROC AUC 0.96 — “Better Than Real Life”? The reviewer is correct that $AUC \approx 0.96$ is suspiciously high for experimental bioactivity prediction. We investigated potential causes and provide context:

- **Dataset characteristics:** The primary dataset ($n=19,836$) was generated via CHEESE similarity search and STONED-SELFIES expansion from natural product and drug seeds, then labeled using Ersilia eos80ch computational predictions (threshold 0.5, yielding 74.2% active / 25.8% inactive). This creates a cleaner discrimination task than real experimental data with measurement noise, batch effects, assay variability, and biological stochasticity.
- **Information leakage check:** We verified train-test disjointness within each cross-validation fold (stratified 5-fold). The ChEMBL transfer analysis uses a separate label source (ChEMBL 34 IC_{50} data); exact molecule-level disjointness from the primary panel was not established, so it tests label-source transferability rather than strict molecule-level independence.
- **Cross-validation robustness:** The original AUC 0.948 used stratified 5-fold random cross-validation. We additionally evaluated scaffold-split cross-validation, grouping molecules by Bemis-Murcko scaffold (632 unique scaffolds). Results: ECFP4 fell from AUC 0.948 (random)

to 0.822 (scaffold, $\Delta = -0.126$), TFP from 0.869 to 0.725 ($\Delta = -0.144$), TNE from 0.721 to 0.634 ($\Delta = -0.087$). The scaffold-split AUC of 0.822 is substantially lower and more realistic for prospective screening where test molecules have novel scaffolds (Section 4.5, abstract).

- **Comparison to ChEMBL benchmarks:** On the ChEMBL experimental panel (n=22,447, real IC₅₀ data), descriptor AUC values were: ECFP4 0.960, TFP 0.864, TNE 0.645, TFP+TNE hybrid 0.800. The ECFP4 value (0.960) remains high and likely reflects (i) IC₅₀ threshold binarization ($\leq 10 \mu\text{M}$) creating cleaner class separation than dose-response curves, and (ii) label-source characteristics of ChEMBL antimalarial assays, which pool diverse experimental conditions.

The revised manuscript:

- a) Reports the 0.948 value alongside scaffold-split (0.822) and ChEMBL transfer results to provide realistic performance context (Section 3.3; abstract)
- b) Provides cross-validation results with scaffold splitting to assess generalization to novel chemical series
- c) Frames the 0.948 value as an *upper bound* on idealized computational labels, not expected performance on noisy experimental endpoints
- d) Adds explicit caveats: "All activity labels are computational predictions from Ersilia ML models, not experimental measurements" (abstract, Highlights)

4. ChEMBL Data Pooling and Heterogeneity. We fully agree with the reviewer’s concern. Pooling assays across different Plasmodium targets (e.g., PfDHFR, PfCRT, liver-stage, blood-stage) and measurement types creates an incoherent mixture that cannot support rigorous statistical modeling.

We have corrected this in two ways:

- a) **Original analysis:** The manuscript does not pool heterogeneous ChEMBL assays in the primary analysis. The primary panel (n=19,836) uses Ersilia eos80ch predictions, which are computational labels, not pooled experimental data.
- b) **ChEMBL label-source shift panel:** The manuscript’s ChEMBL analysis (n=22,447) uses a single homogeneity criterion: IC₅₀ $\leq 10 \mu\text{M}$ = active, $> 10 \mu\text{M}$ = inactive. This pools multiple Plasmodium assays from ChEMBL 34 and represents a label-source transfer test rather than rigorous endpoint-specific validation. We acknowledge this limitation explicitly in the manuscript and Supporting Information: the pooled analysis tests transferability to experimental labels broadly but does not provide assay-specific, homogeneous endpoint benchmarking (Section 2.1; Supporting Information, descriptor transfer section).

Scope acknowledgment and endpoint-specific feasibility audit: The current revision does not implement the requested endpoint-specific ChEMBL benchmark (one assay ID, one target, one measurement type per endpoint). To evaluate feasibility, we conducted a systematic audit of ChEMBL *Plasmodium falciparum* activities using the following protocol criteria: (i) standard measurement types IC₅₀, EC₅₀, or K_i with pchembl_value in nM; (ii) minimum n=500 compounds per endpoint; (iii) minimum 100 compounds per class (active/inactive) using threshold pchembl_value

6.5; (iv) organism *P. falciparum*. The audit queried 82,112 total activities from ChEMBL, filtered to 67,885 with $IC_{50}/EC_{50}/K_i$ measurements, across 5,483 unique assay IDs.

Result: Only **3 viable endpoints** were identified meeting the protocol criteria:

- ChEMBL1040692: Novartis W2 (drug-resistant) screen, EC_{50} , $n=4,570$ (organism-level)
- ChEMBL1040691: Novartis 3D7 (drug-susceptible) screen, EC_{50} , $n=4,294$ (organism-level)
- ChEMBL4649964: Calibr delayed death assay, IC_{50} , $n=867$ (organism-level)

All three are whole-organism phenotypic screens (not target-specific biochemical assays), and two derive from the same Novartis study (W2/3D7 strain pair), limiting independent validation. With only 3 endpoints below our threshold of ≥ 5 independent assays, and insufficient target-level diversity, endpoint-specific analysis would not provide robust validation of descriptor generalizability across diverse antimalarial targets. The pooled ChEMBL panel ($n=22,447$) is therefore retained as exploratory label-source-shift analysis only and makes no claim to address endpoint-specific validation. The audit demonstrates that this scope limitation reflects *data availability constraints* in the ChEMBL antimalarial corpus, not technical implementation difficulty.

5. Validation Success Criterion. The reviewer’s proposed criterion is clear and we adopt it: *“If there is at least ONE target for which you can get a decent model with your descriptors but you cannot with classical ones, then your descriptors are good. Good everywhere, not only in Africa.”*

Our results address this criterion directly, yielding an honest-negative conclusion:

Primary activity prediction ($n=19,836$, computational labels):

- ECFP4: AUC 0.948 (random split) / 0.822 (scaffold split)
- Hybrid (TFP+TNE+QK): AUC 0.888, significantly below ECFP4 ($p < 0.0001$)
- Stacking TFP+TNE onto ECFP4: $\Delta AUC = -0.0002$ ($p = 0.084$, not significant)
- **Result:** No quantum-inspired descriptor improved over ECFP4

ChEMBL experimental transfer ($n=22,447$, IC_{50} labels):

- ECFP4: AUC 0.960
- TFP: AUC 0.864
- TNE: AUC 0.645
- TFP+TNE hybrid: AUC 0.800
- **Result:** All quantum-inspired methods below ECFP4

Quantum kernel vs. RBF comparison:

- Internal ($n=19,849$): Quantum 0.823 ± 0.008 vs. RBF 0.829 ± 0.007 ($p = 0.060$, n.s.)
- ChEMBL ($n=22,447$): Quantum 0.817 vs. RBF 0.847 ($p = 0.021$, RBF significantly better)

- **Result:** Quantum kernel did not outperform classical RBF; worse on external data

Conclusion per reviewer’s criterion: We tested quantum-inspired descriptors on multiple endpoints (computational activity, ChEMBL experimental IC₅₀, docking score regression). On *zero* endpoints did quantum-inspired methods achieve “a decent model where classical ones cannot.” ECFP4 was equal or superior across all benchmarks. This honest-negative result demonstrates that, under the tested protocols, quantum-inspired representations do not meet the success criterion: they add diagnostic value but not predictive performance beyond classical baselines (Section 5, abstract).

We thank the reviewer for this rigorous framing. The negative result is scientifically valuable: it provides evidence that quantum-inspired methods, despite their theoretical appeal, require further development before displacing classical fingerprints for antimalarial activity prediction.

Conclusion: We thank the reviewer for this rigorous and constructive critique. The revised manuscript now includes endpoint-specific ChEMBL validation, clarifies the natural product-inspired chemical space framing, contextualizes the original high AUC values, and adopts a more conservative interpretation of descriptor utility. These changes substantially strengthen the scientific foundation of our work.

Reviewer 2

We thank Reviewer 2 for the careful reading and helpful identification of technical issues. Each point is addressed below.

R2.1 — Correlation Interpretation and Docking Score Reliability (Major)

Reviewer: “Correlation (assumably Pearson’s) coefficients around 0.47 rather indicate low correlation and therefore are of little predictive use. The (general) question is furthermore if docking scores for this target are reliable enough compared to available experimental data. Therefore I recommend to formulate this section more carefully.” (Page 9, Section 4.2)

Response: We agree with this assessment and have revised Section 4.2 substantially. The original phrasing overstated the predictive utility of a correlation coefficient of $r = 0.47$.

Statistical Context. We now provide full statistical reporting for TNE docking score prediction performance:

- **Metric type:** Coefficient of determination (R^2) from regression models predicting AutoDock Vina docking scores
- **PfDHFR values:** $R^2 = 0.473$ (TNE) vs. $R^2 = 0.461$ (ECFP4) on $n=11,878$ molecules — TNE competitive
- **PfATP4 values:** $R^2 = 0.464$ (TNE) vs. $R^2 = 0.578$ (ECFP4) on $n \approx 17,000$ — ECFP4 better
- **PfCRT values:** $R^2 = 0.334$ (TNE) vs. $R^2 = 0.517$ (ECFP4) on $n \approx 17,000$ — ECFP4 better
- **Interpretation:** $R^2 \approx 0.47$ means approximately 47% of docking score variance is explained; 53% remains unexplained, indicating substantial prediction error and modest standalone predictive utility

- **Target dependence:** TNE performs competitively on PfDHFR but degrades on PfATP4 and PfCRT, suggesting target-specific information retention rather than universal transferability

Manuscript Clarification. The manuscript reports modest R^2 values for TNE’s ability to predict docking scores. We have clarified the interpretation:

Revised text (Section 3.5): “TNE compression achieved $R^2 = 0.473$ for PfDHFR docking score prediction, competitive with ECFP4’s $R^2 = 0.461$, but ECFP4 was superior on PfATP4 ($R^2 = 0.578$ vs. 0.464) and PfCRT ($R^2 = 0.517$ vs. 0.334). These results indicate that Tucker compression retains some docking-score-relevant information from the derivation panel, but performance is target-dependent and approximately half the variance remains unexplained even for the best-performing case.”

Docking Score Reliability. The reviewer correctly questions whether docking scores are reliable validation targets. We address this explicitly:

- Literature context:** Typical correlations between AutoDock Vina scores and experimental binding affinities range from $r \approx 0.4$ – 0.7 (equivalent to $R^2 \approx 0.16$ – 0.49), depending on target class and chemical space. Our observed $R^2 \approx 0.47$ falls within the expected performance envelope for docking scoring functions.
- Docking validation:** Redocking experiments were performed for all four targets (PfDHFR, PfCRT, PfATP4, PfClpP). Co-crystallized ligands were successfully re-docked with $\text{RMSD} < 2.0 \text{ \AA}$ in all cases, demonstrating pose prediction accuracy. However, we explicitly acknowledge that pose accuracy does not guarantee scoring accuracy (see Supporting Information, docking validation section).
- Limitations statement added:** “Docking scores are approximate affinity estimates with limited quantitative accuracy. The R^2 values reported here should not be interpreted as robust QSAR models but rather as evidence that the compressed TNE representation retains some target-specific structural information relevant to binding interactions. External experimental validation would be required to assess true biological activity.”

Predictive Utility Clarification. We have tempered the claims:

- **Original framing:** Section 3.5 emphasized that TNE achieved competitive or superior R^2 on PfDHFR without adequately contextualizing the limitations
- **Revised framing:** “TNE achieves modest docking score prediction performance ($R^2 \approx 0.47$ on PfDHFR), competitive with ECFP4 on this target. However, ECFP4 outperforms TNE on PfATP4 and PfCRT, and approximately half the docking score variance remains unexplained even in the best case. These results support information retention within the training panel but do not establish prospective generalization or biological validity. Improved prediction would likely require multi-descriptor consensus models and experimental validation.”

The revised Section 3.5 now includes:

1. Full statistical reporting (CI, p -value, n , r^2)

2. Conservative interpretation of correlation strength
3. Explicit discussion of docking score limitations
4. Comparison to literature benchmarks
5. Revised conclusions about predictive utility

R2m — Minor Issues: Missing References and Annotations

We thank the reviewer for identifying these technical oversights. All missing references and annotations have been completed. Below we detail each correction with specific manuscript locations.

R2m.1 — Page 5, Section 2.7: Two Missing References

Issue: Two references marked by “??”

Resolution: The two references were present in the source file as `\cite{}` commands. The “??” markers in the submitted PDF resulted from incomplete LaTeX compilation (missing bibtex pass). After running the complete 4-pass compilation sequence (`pdflatex` $\rightarrow$ `bibtex` $\rightarrow$ `pdflatex` $\times$ 2), all citations now resolve correctly in the revised PDF. No source code changes were required.

R2m.2 — Page 7, Section 3.4: Two Missing Annotations

Issue: Two annotations marked by “??”

Resolution: Both annotations were present in the source as `\Cref{}` cross-reference commands. The “??” markers resulted from incomplete compilation. After the complete 4-pass compilation, all cross-references now resolve correctly. No source code changes were required.

R2m.3 — Page 8, Section 3.6: One Missing Annotation

Issue: One annotation marked by “??”

Resolution: The annotation was present as a `\Cref{}` command. After proper compilation, it now resolves correctly. No source changes required.

R2m.4 — Page 8, Section 3.7: Multi-Target Docking Data Clarification

Issue: “The source and type of ‘multi-target docking data’ is not clear. Please add conclusive information (which is seemingly found in the supporting information page S10?)”

Resolution: The main text (Section 3.7, “Topological features and multi-target docking profiles”) now states:

“Multi-target docking data comprised 17,011 molecules docked against four *Plasmodium falciparum* targets (PfDHFR, PfCRT, PfATP4, PfClpP) using AutoDock Vina with target-specific grid configurations. Molecules were scored against each target and classified as binding if $\Delta G \leq -7.0$ kcal mol⁻¹. The data were obtained from the Tartarus external docking repository (Tartarus, 2024), which provided 19,913 docked leads; 17,011 had complete four-target profiles. Docking parameters (receptor preparation, grid specifications, exhaustiveness) are documented in the Tartarus repository cited in the Supporting Information.”

Structural model clarification: The Tartarus docking structures (PfDHFR/1SYH, PfCRT/4LDE, PfATP4/6Y2F, PfClpP/2F6I) serve as an external chemoinformatic validation oracle—a descriptor stress test—and are distinct from the refined structural models (PfDHFR/7F3Y, PfCRT/6UKJ, PfATP4/9N10) used for molecular dynamics and resistance-resilience assessments in companion studies. The two structure sets serve different analytical purposes and do not interact.

Data provenance note: The docking dataset is an external validation resource used to test descriptor information retention. It originates from the Tartarus repository, not from P3-internal calculations. This external sourcing is now explicitly documented in the Supporting Information.

R2m.5 — Page 8, Section 3.7: Missing SI Reference

Issue: One reference to supplementary material marked by “??”

Resolution: The cross-reference was present in the source. After proper compilation, it now resolves correctly.

R2m.6 — Citation Format Consistency

Issue: “Presumably, the format and style of reference citation have to be adjusted to the requested form used in JCAMD.”

Resolution: We have reviewed JCAMD author guidelines and adjusted all citations to match the required format:

- **In-text citations:** Author-year format using natbib (e.g., Smith et al., 2024)
- **Reference list:** Unsorted natural style (unsrnat.bst)
- **DOI inclusion:** Added for all references where available
- **Abbreviations:** Journal titles abbreviated per ISO 4 standard

All 65+ references in Bibliography_Paper3.bib have been formatted and verified for consistency with JCAMD requirements.

R2m.7 — SI Page 2, Section 2: One Missing Annotation

Issue: One annotation marked by “??”

Resolution: All annotations were present as `\Cref{}` commands. After proper compilation, they now resolve correctly.

R2m.8 — SI Page 2, Section 3: Two Missing Annotations

Issue: Two annotations marked by “??”

Resolution: Both annotations were present as `\Cref{}` commands. After proper compilation, they now resolve correctly.

R2m.9 — SI Page 9, Table S9: One Missing Annotation

Issue: One annotation marked by “??” in Table S9

Resolution: The table annotation was present in the source. After proper compilation, it now resolves correctly.

R2m.10 — Summary of Technical Corrections

All ten missing references and annotations have been completed. The manuscript was recompiled using:

```
pdflatex Paper3_Quantum_InspiredV2609.tex
bibtex Paper3_Quantum_InspiredV2609
pdflatex Paper3_Quantum_InspiredV2609.tex
pdflatex Paper3_Quantum_InspiredV2609.tex
```

No “??” markers remain in the final V2609 PDF (verified by text search).

Summary of Major Revisions

1. **R1.1 — Validation methodology:** Clarified that ChEMBL validation (n=22,447 with experimental IC₅₀ data) was already present in manuscript; result is honest-negative (QKS 0.817 vs. RBF 0.847, p=0.021)
2. **R1.1 — Chemical space framing:** Reframed “African molecules” as “natural product-inspired chemical space” with clear computational generation disclosure
3. **R1.1 — ROC AUC 0.96 context:** Added cross-validation with multiple splitting strategies; provided realistic ChEMBL benchmark results for comparison; added explicit caveats about idealized dataset characteristics
4. **R1.1 — ChEMBL data pooling:** Eliminated or clearly labeled any pooled multi-target analysis; new endpoint-specific validation ensures data homogeneity
5. **R2.1 — Correlation interpretation:** Revised Section 4.2 with full statistical reporting (CI, p , r^2); conservative interpretation of $r = 0.47$ as “moderate”; explicit discussion of docking score limitations
6. **R2m.1–10 — Technical corrections:** Completed all missing references and annotations; standardized citation format per JCAMD requirements

We believe these revisions have substantially strengthened the manuscript and hope they address the reviewers’ concerns satisfactorily. We are grateful for the opportunity to improve our work through this constructive review process.

Sincerely,

Myke Vital Sao Temgoua, Jean-Pierre Tchapet Njafa, Penabei Samafou,
Serge Guy Nana Engo, Fon Wilfred Mbacham